

The Equine Joint Biologics Landscape at a Glance

How equicenta CTM compares against Arthramid, Pro-Stride APS, ProVet APC, IRAP II, and Adequan i.m.

KEY REFERENCE

Bertone et al. 2026, *Frontiers in Vet Sci* — DOI 10.3389/fvets.2026.1833933. Pragmatic multicenter controlled trial: 138 horses / 195 conditions / 8 U.S. practices / 26-week follow-up.

	equicenta® CTM	Arthramid® Vet	Pro-Stride® APS	ProVet® APC	IRAP® II	Adequan® i.m.
	Cryopreserved equine UC tissue matrix	2.5% polyacrylamide hydrogel	Autologous protein solution (APS)	Autologous platelet concentrate (L-PRP)	Autologous conditioned serum (ACS)	PSGAG — FDA-approved IM DMOAD
Product class & origin What it is	Allogeneic biologic tissue matrix — standardized, off-the-shelf, closed-herd source	Synthetic device — 2.5% polyacrylamide hydrogel	Autologous device — horse's own blood → protein solution	Autologous device — horse's own blood → leukocyte-rich PRP	Autologous conditioned serum — horse's own blood, 24-hr incubation product	FDA-approved drug (PSGAG / DMOAD) — semi-synthetic
Route How it's delivered	Local (intra-articular; peri-tendinous; peri-ligamentous)	Intra-articular only — integrates into synovial membrane	Intra-articular	Intra-articular; soft-tissue at vet discretion	Intra-articular	Intramuscular (systemic; not IA)
Primary MOA What it does	Supports tissue repair & modulates inflammation via ECM proteins, growth factors, EVs	Mechanical & biointegrative — cushions and integrates in synovial membrane	Concentrated IL-1Ra, A2M, growth factors — cytokine blockade	Elevated IL-1Ra:IL-1β ratio & TGF-β1 + platelet growth factors	IL-1Ra-dominant ACS — IL-1Ra competitive antagonism of IL-1β	DMOAD — inhibits catabolic enzymes, stimulates proteoglycan & HA synthesis
Tissue scope in evidence Joints / tendon / ligament	Joints + tendons + ligaments in one multicenter trial	Joint OA only	Joint OA only (anchor RCT single joint)	Characterization only; use spans joints & soft tissue	Joint OA only	Joint OA only (labeled: carpal & hock)
Anchor peer-reviewed evidence Strongest publication	Bertone 2026 pragmatic multicenter controlled trial (138 horses / 195 conditions / 8 practices)	de Clifford 2021 JEVS — single-arm proof-of-concept (n≈32)	Bertone & Ishihara 2014 AJVR saline-controlled RCT (n=40, single joint, 14-day endpoint)	Am J Vet Res 2024 characterization only (n=6 horses)	2023 systematic review (PMC10747612) — only 1 of 6 clinical studies had a control group; no definitive efficacy	FDA FOI Summary NADA 140-901 — placebo-controlled adjuvant-carpitis dose-titration (n=36 healthy horses)
Comparator in that study Who it was measured against	Clinician-selected standard of care (real-world active comparator)	No comparator (single-arm)	Saline	No efficacy trial (characterization only)	Variable — most published clinical work lacks a control arm	Saline placebo in an induced adjuvant-carpitis model
Population studied Real-world vs. induced	Naturally occurring disease — performance horses, 8 U.S. specialty practices	Client-owned OA (single-arm)	Client-owned OA	Healthy horses (characterization)	>30k humans / >3k horses reported clinically; controlled data limited	Healthy Quarter Horses & Thoroughbreds with experimentally induced carpalitis
Key outcome signal The number vets will remember	Return-to-performance 52.9% vs. 29.3% at 26 wk (OR 2.72; p<0.006)	Sustained lameness reduction reported in single-arm follow-up cohorts	APS improved lameness & kinetic gait vs. saline at 14 d; response concentrated in mild cases (lameness <4, min. radiographic OA)	N/A — no efficacy trial	Systematic review: no definitive efficacy evidence; individual studies positive but uncontrolled	71% improvement in max carpal flexion after full 7-dose regimen
Onset & durability Timing profile	Separation from SOC at wk 2–4; sustained through 12 wk; performance signal durable at 26 weeks	Effects reported to build over weeks; commercial materials cite months of durability	14-day primary endpoint; commercial materials cite up to 1 year	General PRP experience: onset 1–4 wk; durability variable	Series of 2–3 IA injections over weeks typical in practice	Effects build across 28-day 1/4/7 course; maintenance dosing common
Workflow & equipment Chair time / capex	Off-the-shelf, single-dose, no equipment	In-clinic IA injection; no processing; no capex	~20 min 2-step processing on Zoetis Universal Centrifuge + disposables	<3 min spin on 4.4-lb portable ProVet centrifuge + disposables	24-hr incubation + centrifuge (multi-visit); IRAP II kit + incubator	1/4/7 regimen: 500 mg IM q4d × 7 doses over 28 days; no capex
Dose consistency Batch vs. patient-driven	Batch-tested standardized proteomic + EV profile	Synthetic — manufactured to specification	Varies with donor horse's blood composition	Varies with donor platelet & WBC counts	Varies with donor blood + incubation performance	FDA-approved lot-to-lot chemistry
Regulatory status (U.S.) How each is positioned	Working with FDA CVM to open NADA; cGTP → cGMP by end of Q3	Cleared/marketed as a device	Cleared/marketed as an autologous device	Cleared as a medical device; IFU: efficacy “not established” for bone healing, hemostasis, or other clinical use	Marketed as an autologous ACS system	FDA-approved drug, NADA #140-901; Rx-only

QUICK CASE-SELECTION GUIDE — LEAD WITH equicenta® CTM WHEN...

Case involves tendon or ligament alongside joint disease — most competitors have joint-only evidence; eCTM has published outcomes across joints, tendons, and ligaments in one multicenter trial.

Practice wants peer-reviewed clinical efficacy against a real-world active comparator (not a single-arm study, not a saline-controlled 14-day endpoint, not a characterization study).

Older, metabolic, or heavily pre-treated horse where autologous quality (APS / APC / IRAP) is likely to be variable — a standardized allograft removes patient-side variability.

Advanced or radiographically significant OA — the Bertone-Ishihara 2014 APS response was concentrated in horses with lameness <4 and minimal radiographic OA.

Practice doesn't want to invest in a centrifuge or incubation platform — eCTM is off-the-shelf, single-dose, no equipment.

Client wants a single episode of care rather than a multi-visit series (IRAP) or a repeating 7-dose systemic course (Adequan).

Horse is already on maintenance Adequan and still symptomatic — add eCTM at the site of active pathology; Adequan and eCTM are frequently used together.

Off-label joints (stifle, fetlock, coffin) — Adequan's label is carpal & hock only; eCTM was studied across a broader case spectrum.

Sources. equicenta CTM: Bertone et al. 2026, *Frontiers in Vet Sci* (DOI 10.3389/fvets.2026.1833933); BMC Medicine 2025 characterization & 42-day biocompatibility; ACVSMR 2026 EV/exosome poster. Arthramid Vet: de Clifford et al. 2021, *Journal of Equine Veterinary Science*. Pro-Stride APS: Zoetis product page; Bertone & Ishihara 2014, *AJVR*; Usimaki/Watkins 2024, *EVJ*. ProVet APC: Dechra ProVet APC IFU; Am J Vet Res 2024 characterization. IRAP II: Vet Sci 2023 systematic review (PMC10747612); Arthrex IRAP scientific update; Hraha et al. 2011, *EVJ* (PubMed 21496084). Adequan i.m.: *DailyMed* label; FDA FOI Summary NADA 140-901. Trademarks & disclaimer. equicenta® CTM: Equine Performance Labs. Arthramid®: Cuntura International A/S. Pro-Stride®, Restigen®, Zoetis®: Zoetis Inc. ProVet®, Dechra®: Dechra Pharmaceuticals PLC. IRAP®: Orthogen AG / Arthrex, Inc. Adequan®: American Regent, Inc. (Luitpold Pharmaceuticals). Comparator information is provided for reference under fair-use vet-education purposes and does not imply endorsement; the Bertone 2026 trial did not evaluate these products as named head-to-head comparator arms. Product selection is a clinical decision for the treating veterinarian; individual results may vary.